P R O J E C T C H R O N I C L E

UMUD Muscle Architecture Challenge

A Complete Build Chronicle: From 1.114 to 0.510

Domain: Quantitative musculoskeletal ultrasound image analysis

Targets: Pennation angle / fascicle length / muscle thickness

Outcome: 0.510 leaderboard score, from a 1.114 constants baseline

Scope: 20 submissions, 4 model generations, 15 documented failures

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Executive Summary

This document is the complete, chronological record of our campaign on the UMUD muscle architecture prediction challenge. It covers every phase of the work exactly as it unfolded across our conversations: the initial reading of the task, the forensic analysis of the data, the strategic decision that fixed the architecture, the evolution of four model generations, the construction of a validation harness, the calibration detective work, all twenty leaderboard submissions, and every experiment that failed along the way. It is written so that a reader with the original data and this document alone could reproduce the final result, and so that the reasoning behind each decision survives long after the leaderboard closes.

The task was to predict three muscle architecture parameters from a single 2D B-mode ultrasound image: pennation angle (`pa_deg`), fascicle length (`fl_mm`), and muscle thickness (`mt_mm`). The composite score is the mean of per-target errors normalized by their tolerances of 6 degrees, 12 mm, and 3 mm respectively, so lower is better. Our first submission was a deliberately naive constants baseline that scored **1.114** and established the floor. Our best submission, version 12 with a 0.95 calibration multiplier (`v12_m5`), scored **0.510**, a reduction of roughly 54 percent, placing the project in the neighborhood of 130th to 160th of more than 328 entrants. Every intermediate step between those two numbers is documented in Chapter 9.

One discovery shaped the entire project more than any other: **the training data contains no regression labels at all**. There are no pennation angles, no fascicle lengths, no thickness values anywhere in the training set. What exists is two disjoint pools of segmentation masks, one annotating aponeurosis bands and one annotating fascicle lines, and the two pools share zero images. Every predicted number therefore had to be derived geometrically from predicted masks and then converted from pixels to millimetres through per-image calibration. This single constraint dictated a three-stage architecture, segmentation, geometry, calibration, that never changed from the day it was chosen.

The largest score movements came not from model upgrades but from understanding the data. Fixing the calibration of the PNG test family was worth 0.19 points. Correcting pennation angle for measured shrinkage was worth another 0.057. Discovering, through a four-submission bisection, that the tif-family calibration constants were five percent too high was worth 0.066 and produced the final best score. The model generations contributed steadily but more modestly, and the last third of the campaign demonstrated that post-processing has a ceiling: after calibration and correction were exhausted, three successive model-side ideas all failed on the leaderboard and the best configuration remained unchanged.

The document is organized chronologically within a thematic frame. Chapters 1 through 3 cover the problem, the data forensics, and the strategy decision. Chapters 4 and 5 describe the models and the geometry engine. Chapters 6 through 8 cover validation, calibration, and post-processing. Chapter 9 is the complete submission log with the score trajectory. Chapter 10 catalogs the negative results. Chapters 11 through 14 document the final architecture, the infrastructure lessons, the unbuilt roadmap, and the meta-lessons, followed by two appendices listing every artifact and every constant.

1. The Problem and the Ground Rules

1.1 The task in plain language

Muscle architecture refers to the arrangement of muscle fascicles within a muscle, and it is the primary determinant of how a muscle trades force against velocity. Ultrasound is the standard non-invasive way to measure it: a B-mode probe images a longitudinal plane of the muscle, and the visible striations are the fascicles inserting into the bright collagen sheets called aponeuroses. The challenge asked for three quantities per image, and the meaning of each is worth stating precisely because the geometry engine in Chapter 5 is built directly on these definitions.

Table 1: The three prediction targets and their scoring tolerances

Target	Definition	Unit	Tolerance
`pa_deg`	Pennation angle: the angle between a fascicle and the deep aponeurosis it inserts into	degrees	6
`fl_mm`	Fascicle length: the distance along the fascicle between the superficial and deep aponeuroses, extrapolated when the fascicle exits the field of view	mm	12
`mt_mm`	Muscle thickness: the perpendicular distance between the two aponeuroses, measured at three sites and averaged	mm	3

The tolerances deserve attention because they are wildly different in relative terms. A 3 mm tolerance on a muscle thickness that typically ranges between 15 and 35 mm is tight; a 12 mm tolerance on fascicle lengths that can reach 100 mm or more is generous. The composite score, `mean(MAE_PA/6 + MAE_FL/12 + MAE_MT/3)`, therefore punishes thickness errors roughly four times harder than fascicle-length errors per unit of relative error. This asymmetry pushed us toward careful calibration work, since a scale error hits MT and FL simultaneously, and MT is where it hurts most.

1.2 The metric, the split, and the submission budget

Before writing any model code we obtained and verified the official scoring implementation locally. This early step removed all ambiguity about what was being measured, and it let us score candidate submissions exactly as the platform would. Two facts from the rules mattered operationally: the 309 test images are split 33 percent public and 67 percent private, and each account receives five submissions per day. The public-private split means the leaderboard number is itself a noisy estimate of the private score, so small improvements can be indistinguishable from noise, and the daily budget means every submission is a scarce resource that should be spent on a pre-declared hypothesis.

These constraints set the operating doctrine for the whole campaign. First, no submission went to the leaderboard without a local reason to believe it would help, ideally backed by the twins harness described in Chapter 6. Second, risky changes were always paired against a control in a single-variable A/B whenever the budget allowed, so that a score movement could be attributed to exactly one cause. Third, the constants baseline of submission one was treated as a permanent anchor: any configuration that ever scored close to it was considered broken rather than merely unlucky.

1.3 The reference protocol and the missing labels

The organizer's reference protocol described how the ground-truth values were produced: two human raters each traced three fascicles, three pennation angles, and three thickness sites per image, and the reported value is the average. Knowing this mattered twice later in the project. It told us the targets are central tendencies over multiple fascicles rather than the property of a single fascicle, which informed the decision to average the top three selected fascicle components rather than take a single best line. It also told us the tolerance bands were chosen relative to inter-rater variability, meaning a perfect model cannot reach zero error, and the realistic floor is the raters' own disagreement.

Then came the discovery that redefined the project. A careful pass through every folder of the training data confirmed that **there are no regression labels anywhere**. The training material consists exclusively of images paired with binary segmentation masks, in two pools: 1,048 images with aponeurosis masks and 2,761 images with fascicle masks. Not a single image carries both mask types, and no file contains a measured angle, length, or thickness. Any approach that trains a regressor directly on images therefore has nothing to learn from. Every number we ever submitted was computed by running segmentation models, extracting geometry from the predicted masks, and converting pixels to millimetres through per-image calibration. The entire architecture of the solution follows from this one fact, and it is why the project spent its first phase not on models but on data forensics.

2. Data Forensics: Reading the Data Before Building Anything

2.1 The lay of the land

The data arrived in a `Data/` directory with a deceptively simple top level: two image folders, two mask folders, and a test folder. Reality was messier. The folders were nested at irregular depths, several names contained typos, and the apparent structure could not be trusted. The first engineering decision was therefore to ignore the directory tree entirely and auto-discover every file by path substring matching, so that `apo_imgs_v1`, `fasc_masks_v1`, and `test_images_v2` were located by signature rather than by position. This avoided an entire class of silent failure where a renamed folder causes a pool to be quietly skipped.

Table 2: Data inventory as discovered by the forensics pass

Pool	Content	Files
`apo_imgs_v1`	B-mode ultrasound images with aponeurosis annotations	1,048
`apo_masks_v1`	Binary aponeurosis band masks for the above	1,048
`fasc_imgs_v1`	B-mode ultrasound images with fascicle annotations	2,761
`fasc_masks_v1`	Binary fascicle line masks for the above	2,761
`test_images_v2`	Evaluation images, no masks provided	309

The forensics phase produced a set of standing artifacts that the rest of the project consumed: `content_hashes.csv` recording the cryptographic and perceptual identity of every image, `twins.csv` cataloguing the matched pairs described below, `fasc_mask_inventory.csv` with per-mask component statistics, and `calibration_v1.csv` mapping each test image to its calibration constants. Everything in Chapters 4 through 9 depends on these files, which is why the forensics work is documented here in detail rather than summarized away.

2.2 The hash investigation: one thousand names that lied

The most dangerous trap in the dataset hid in a filename statistic. Exactly 1,048 filenames appeared in both the aponeurosis pool and the fascicle pool, which naturally suggested that these were the same images annotated twice, once per structure. Had that been true, the pipeline could have been trivial: a single set of images with both mask types, perfect for training a joint model and for validating geometry end to end. Before building on that assumption we hashed the actual pixel content of every image, and the assumption collapsed.

The content hashes proved that the two pools share **zero images**. Counting unique content across the whole training set gave 2,091 unique images in the fascicle pool and 1,029 in the aponeurosis pool, totalling 3,120 unique images, with no overlap whatsoever. The 1,048 matching filenames were pure name collision produced by two independent indexing systems that happened to use overlapping numeric ranges. The practical consequence was

severe: no image anywhere has both an aponeurosis mask and a fascicle mask, so the final pipeline would always have to predict aponeuroses on images whose aponeurosis ground truth it could never check directly, and the two model families would have to generalize across pools trained on only one structure each.

2.3 The 122 twins: an accidental gold mine

While running the content-hash comparison we searched not just for identical names but for identical pixels across pools, and this time the search hit. There are **122 fascicle-pool images with a byte-identical counterpart in the aponeurosis pool**: correlation 1.0000, mean pixel difference 0.09, identical in every respect that matters. Each of these images, which we called twins, therefore has both a real fascicle mask and a real aponeurosis mask on identical image content. They are the only images in the entire project with complete ground truth for both structures.

The twins immediately became the project's validation set and its most valuable asset. Both model families exclude them from training, which keeps them honest, and the full prediction chain, both segmentation models plus geometry plus calibration, can be run on them and scored against geometry computed from the real masks. Chapter 6 describes the harness built around them. It is not an exaggeration to say that without the twins we would have been flying blind on a five-submissions-per-day budget; with them, nearly every leaderboard verdict was predicted in advance, at zero submission cost.

2.4 Duplicates and the two annotation conventions

The hash pass also exposed two duplication problems. In the fascicle pool there were **670 exact duplicate image-and-mask pairs**, confirmed by the organizers as a merge bug on their side. Training on them would have triple-counted 670 images, biased validation splits, and inflated confidence in nothing useful, so they were deduplicated before any training run. In the aponeurosis pool there were 19 pairs of duplicate images with **different masks**, and inspection showed why: the two masks alternated between two annotation conventions, one drawing the aponeurosis band wider and one narrower, apparently depending on which rater or session produced them. Since there was no way to decide which convention the test ground truth followed, all 38 files were dropped rather than risk teaching the model a coin flip.

2.5 Mask semantics and the resolution mismatch

Understanding what the masks actually encode was its own small investigation. All masks are binary with values 0 and 255. Fascicle masks are extremely sparse: roughly 0.2 percent of pixels, forming 1-pixel-thick traced lines that follow individual fascicles across the image. Aponeurosis masks are thick bands covering roughly 2 to 4 percent of pixels. This difference

in kind, thin lines versus thick bands, is why the two structures always had separate models with separate target treatments, and why the fascicle targets needed dilation during training to give the network a learnable signal at all.

A separate problem was that many image and mask pairs disagreed on pixel dimensions. Three interpretations were possible: the mask could be a crop, a pad, or a plain stretch of the annotation. We resolved the question empirically with intensity-contrast tests and by overlaying masks on twin images, and the answer was unambiguous: masks relate to images by **plain bilinear stretch**. This finding simplified every downstream component, since a predicted mask at model resolution can be stretched back to image coordinates with a single resize, with no crop offsets to track.

2.6 Video sequences, clusters, and bursts

Ultrasound is a video modality, and the static frames in the dataset did not escape that fact. By computing 32-pixel thumbnails for every image and clustering with union-find at a correlation threshold above 0.99, we found sequence structure everywhere: training contains contiguous video runs, with the largest single cluster at 576 frames. The test set is structured the same way, containing **28 bursts totalling 138 images** where consecutive frames are near-duplicates of each other.

Two consequences followed. On the training side, apparent dataset size overstates effective diversity, which mattered when reasoning about augmentation and fold construction. On the test side, near-duplicate frames mean per-image predictions within a burst are highly correlated measurements of the same underlying anatomy, so **averaging predictions across a burst is legitimate variance reduction rather than leakage**. Burst averaging entered the final pipeline and survived every A/B it was tested in.

2.7 The test muscle map

The 309 test images are not a homogeneous sample. Each carries a burned-in device interface, and clustering the interface signatures identified **29 distinct UI-layout families**. With predictions in hand, cross-family prediction tables revealed which families belonged to which muscle. Families 0 and 22 behave like vastus lateralis, the classic unipennate textbook case, with moderate pennation angles and moderate lengths. Families 20, 21, and 24 behave like **rectus femoris**: long fascicles, low pennation angles, thick muscles, and, crucially, a bipennate architecture with fascicles fanning in opposite directions from a central aponeurosis. The PNG family behaves like gastrocnemius, with steeper angles. At least one family comes from a Philips Lumify handheld device that never appears in training, a pure domain-shift case.

This muscle map earned its keep twice. Early on, it calibrated expectations: predictions of long fascicles for families 20 to 24 were confirmed as correct anatomy rather than flagged as model failure. Late in the campaign, it explained the catastrophic failure of a classical structure-tensor blend, which collapsed precisely on the bipennate rectus femoris families where opposing fascicle fans cancel each other in a circular mean. That story is told in Chapter 10.

2.8 The one percent warning

The last forensic finding was a quiet one that haunted the entire campaign. Measuring the angle of every annotated fascicle component in the training masks showed that only **1.1 percent exceed 25 degrees**. The training data overwhelmingly shows shallow-angle fascicles, which means any learned model inherits a steep-angle blind spot, and any geometry filter tuned on this distribution will quietly reject the steep fascicles that do occur in test families like the gastrocnemius-like PNG group. Every harness run later confirmed the blind spot was real. The practical responses were the 6-degree floor and 60-degree ceiling on candidate acceptance angles and the selection of fascicles by dominant mode rather than by maximum angle, both described in Chapter 5, but the underlying data imbalance was never fully cured.

3. The Strategy Decision: Why Segmentation Instead of Regression

With forensics complete, the architecture question presented itself. The default instinct for a prediction competition is an end-to-end regressor, an ImageNet backbone with three output heads, but Section 1.3 had closed that door: there are no target values to regress against. The options that remained were all variations of deriving the numbers ourselves, and the choice among them was the most consequential decision of the project.

The alternative we rejected first was training a regressor on pseudo-labels, that is, on values computed by some initial pipeline and then fed back as targets. This is circular in the absence of any external anchor: the regressor can at best imitate the initial pipeline, and Chapter 4's distillation experiment later confirmed empirically that imitation of a weak pipeline does not improve on it. The second alternative, classical image processing end to end, was rejected on robustness grounds: speckle noise and the huge appearance diversity across 29 device families make hand-crafted line detection fragile precisely on the domain-shift families where errors were already largest.

The chosen architecture decomposes the problem into three stages with clean interfaces. **Stage one is segmentation**: two specialized U-Net models, one for aponeurosis bands and one for fascicle lines, trained on the respective mask pools. Segmentation transfers across

domains far better than global regression because it is a local, per-pixel task, and the mask pools give it thousands of genuine supervised examples. **Stage two is geometry**: a deterministic module that converts a pair of aponeurosis lines and a set of fascicle lines into pennation angle, fascicle length in pixels, and thickness in pixels, using the anatomical definitions of Table 1 directly. **Stage three is calibration**: converting pixels to millimetres per image using the burned-in depth rulers of the device interfaces.

Two design choices inside this decomposition are worth defending explicitly. First, we used **two separate networks** rather than one multi-class segmenter, because aponeuroses and fascicles differ in kind, thick bright bands versus faint one-pixel lines, and separate models permit separate resolutions, target dilations, and thresholds. The cost is twice the inference code; the benefit is that each structure gets the treatment it needs, and a failure in one family does not poison the other. Second, every stage is **individually testable**: the segmentation stage is scored against the twins, the geometry stage is scored against real masks, and the calibration stage is scored against the leaderboard. When something broke, we always knew which stage to blame.

The final piece of the strategy was a **fallback chain** philosophy: no single component failure should ever be allowed to emit garbage. The inference script tries the strongest configuration first, the two FascNet v3 folds ensembled, and falls back through progressively simpler configurations, FascNet v1 with test-time augmentation, FascNet v1 alone, and finally the constants 18.0 degrees, 80.0 mm, and 23.0 mm, if anything in the chain fails. The constants floor also served as submission one, anchoring the leaderboard at 1.114 and proving the submission format end to end before any model output existed.

Hindsight validated the strategy decision completely. Across the entire campaign, through four model generations and twenty submissions, the three-stage decomposition never once needed to change. Every failure in the second half of the project was a wrong constant or a wrong post-processing assumption inside the stages, never a failure of the stage boundaries themselves. The architecture was, in the best sense, boring, and that is exactly what let the interesting work, the calibration detective story of Chapter 7, proceed on a stable foundation.

4. The Model Journey, Model by Model

This chapter records the model lineage in the order it was built, including the reasoning that produced each generation and the evidence that retired the previous one. Four generations of fascicle segmenter were trained, alongside one aponeurosis segmenter, one distillation regressor, and two ensemble variants that never shipped. Table 3 summarizes the inventory; the sections that follow tell the story.

Table 3: Model inventory and final status

Model	Resolution	Targets	Final status and role
`ApoNet v1`	512	apo bands	Production aponeurosis model; cross-pool twin dice 0.82
`FascNet v1`	512	fasc, dilation-2	Retired from production; kept as retry-chain safety net
`FascNet v2` (3 folds)	512	fasc, dilation-2	Superseded by v3; source of the +110 px FL-bias diagnosis
`FascNet v3` (folds 0, 1)	640	fasc, dilation-1	Production fascicle models; twin FL bias +37 px
`FascNet v3` (fold 2)	640	fasc, dilation-1	Collapsed twice; permanently dropped
`ApoNet` 2-seed ensemble	512	apo bands	Failed on leaderboard (minus 0.107); removed
`regressor_v1` (ResNet50)	224	PA/FL/MT heads	Distillation experiment; rejected, never submitted

4.1 ApoNet v1: betting the pipeline on cross-pool transfer

ApoNet came first because everything else depended on it. The geometry engine needs both aponeurosis lines on every image, but aponeurosis masks exist only in the aponeurosis pool, and Section 2.2 had proven the pools are disjoint. So the very first model carried the existential risk of the whole strategy: a U-Net with a ResNet34 encoder, trained on the 888 clean unique aponeurosis images (after deduplication, removal of the 19 ambiguous pairs, and exclusion of the twins), had to segment aponeuroses on fascicle-pool-style and test-style images it had never seen with aponeurosis labels.

Training was deliberately simple and robust: 512 by 512 inputs, a combined BCE-plus-Dice loss, AdamW at 3e-4 with cosine decay, 40 epochs, heavy but conventional augmentation. The design bet was that segmentation is local enough to transfer across acquisition domains, and the bet paid off. On the 122 held-out twins, ApoNet v1 achieved a **Dice coefficient of 0.82 against real aponeurosis masks on content it never trained on**. That number, more than any leaderboard score, is what legitimized the entire three-stage architecture; if cross-pool transfer had failed here, the project would have had to change course fundamentally. ApoNet v1 remained in production, unchanged, through every subsequent submission.

4.2 FascNet v1: the first fascicle segmenter

FascNet v1 was the corresponding model for fascicle lines, trained on the deduplicated fascicle pool with dilation-2 targets, meaning the one-pixel traced lines were dilated by two pixels before use as training targets. The dilation is necessary because a one-pixel target

gives vanishingly sparse positive signal, but it has a cost that only became measurable later: dilated targets bias the learned lines outward, and the bias propagates into every geometric quantity. At the time, v1 did its job, it produced the first end-to-end pipeline and submissions two through four, and it was permanently retained as the safety net at the bottom of the fallback chain, where its well-understood behavior was worth more than a marginal accuracy gain.

4.3 FascNet v2: the dilation experiments and the +110 px diagnosis

FascNet v2 was trained as a three-fold ensemble at 512 resolution with the same dilation-2 targets but improved schedules, and it powered the ensemble-plus-TTA submission v4, which delivered a marginal improvement of about 0.023. Its real contribution, though, was diagnostic. Running the twins harness against v2 models revealed a systematic **fascicle-length overestimation of roughly +110 pixels**, and overlay inspection traced the cause to the dilation: when the network learns to fire on a band around the true line, the extracted skeleton sits systematically off the true fascicle path, and both intersection points with the aponeuroses shift outward.

This diagnosis reframed the problem. The error was not noise to be averaged away; it was a deterministic bias with an identifiable physical cause in the target construction. That insight set up the v3 generation: if dilated targets bias the line outward, the fix is thinner targets, and if thin targets starve the loss, the fix is higher resolution to compensate. v2 taught us what to change; v3 changed it.

4.4 FascNet v3: the resolution and dilation breakthrough

FascNet v3 combined both corrections: **640 by 640 resolution with dilation-1 targets**, 50 epochs at batch size 8. The effect on the twins harness was the single largest model-level improvement of the campaign: the fascicle-length bias fell from +110 pixels to **+37 pixels**, and pennation angle accuracy improved in step. The thinner targets produce truer lines at the cost of a harder learning problem, and the higher resolution pays that cost by giving the network more pixels to resolve each fascicle.

v3 also delivered the most counterintuitive infrastructure lesson of the project: **each model must predict at exactly its own training resolution**. During v3 integration, an early evaluation ran the 640-trained models at 512 inputs for convenience, and the results looked plausible while being silently wrong, because the confidence calibration of a segmentation network is tied to the resolution it trained at. The rule that entered the codebase, and the one we would flag to anyone reusing these checkpoints, is that v3 predicts at 640, v1 and ApoNet predict at 512, and the ensemble averages probabilities only after each member has predicted at its native resolution. Following this rule, the two healthy v3 folds probability-averaged became the production fascicle segmenter from submission v11 onward.

4.5 The fold-2 collapse and the ensemble lessons

The plan was three v3 folds; reality delivered two. Fold 2 collapsed not once but twice, in both cases presenting as an early-schedule spike in loss followed by convergence to a degenerate solution. Retraining reproduced the failure, which pointed to an unlucky interaction of seed and data order rather than a code bug. A tempting option was to salvage mid-schedule checkpoints from before the spike, but inspection showed those checkpoints carried abnormally spiky probability maps, and probability-averaging a poisoned member into an ensemble degrades the average even when the member's Dice looks acceptable. Fold 2 was dropped permanently, and production ran on folds 0 and 1.

The episode generalized into two standing rules. First, **hot mid-schedule checkpoints are never ensemble members**: a checkpoint from an unstable region of training can poison a probability average in ways that a single-model metric will not reveal. Second, **correlated folds are not diversity**: folds trained on near-identical data with identical hyperparameters produce near-identical models, and averaging them buys almost nothing, which is exactly what the v2 three-fold ensemble's marginal 0.023 had already hinted. Genuine diversity has to come from different resolutions, different targets, or different architectures, not from reseeding the same recipe.

4.6 The dead-end branches

Four model-side ideas were built and rejected, and each rejection taught something permanent. **Rotation augmentation at plus-or-minus 30 degrees**, trained for submission v4's second attempt, degraded every twins metric, collapsing correlation from 0.60 to 0.38; the lesson is that augmentation must match the test distribution, and test images are upright by construction, so wide rotations teach the network to find fascicles in orientations that never occur. The **ApoNet two-seed ensemble** was the most expensive rejection: it looked safe by every local measure, then cost 0.107 on the leaderboard in submission v15, establishing that a probability-averaged consensus of two segmentation models can behave differently from either member in the downstream geometry, and that segmentation ensembles must be evaluated end to end, never assumed.

The **distillation regressor** was the most instructive failure. A ResNet50 with three regression heads was trained to imitate the full pipeline's outputs on pseudo-labels generated from training images. It fit those pseudo-labels beautifully, 0.87 degrees of PA error against them, and then produced 4.29 degrees against the twins, several times worse than the pipeline it was imitating. Imitating a teacher is not the same as solving the task: the regressor learned the teacher's central tendency and discarded the per-image evidence, exactly the failure mode that local, per-pixel segmentation avoids. Finally, **fine-tuning on test-image pseudo-labels** was evaluated and rejected on paper without a run, on the argument that it would reinforce regression-to-the-mean precisely on the domain-shifted images where the pipeline was

weakest, and an **anisotropy correction** for fascicle length, scaling by 0.85 to hypothesized vertical stretch, tested neutral and was dropped once the raters' isotropic protocol made its premise moot.

5. The Geometry Engine

Between the segmentation masks and the three predicted numbers sits roughly four hundred lines of deterministic geometry. It is the least glamorous and most iterated code in the project: every filter threshold in it was the subject of at least one A/B, and several plausible-sounding upgrades to it made things worse. This chapter documents the version that shipped, followed by the ideas that failed against it. The engine runs identically on predicted masks in production and on real masks in the twins harness, which is what makes the harness meaningful.

5.1 Aponeurosis extraction

The engine takes the two largest connected components of the aponeurosis mask, requiring at least 500 pixels each under 8-connectivity, and requires their centroids to sit at least 60 pixels apart to prevent a single fragmented band from being mistaken for two. The component with the higher centroid is the superficial aponeurosis and the lower one is the deep aponeurosis, matching the anatomical convention that thickness is measured between them. Each component is then collapsed to a first-degree polynomial fit, `polyfit`, over its pixel coordinates, which yields two infinite lines that the fascicle intersection math can use directly. The polynomial fit is deliberately simple, and Section 5.5 records why a more sophisticated robust fit was tried and reverted.

5.2 Fascicle candidate filtering

Fascicle candidates are connected components of the fascicle mask that pass four gates. They must contain at least 80 pixels, filtering speckle and fragmentation. Their angle against the deep aponeurosis line must fall between **6 and 60 degrees**, the floor excluding near-parallel junk that hugs the aponeurosis and the ceiling excluding vertical artifacts and the steep-angle noise the model is blind to anyway, per Section 2.8. Their elongation, the ratio of the component's long axis to its short axis, must be at least 4, which keeps compact blobs out regardless of their angle. And their centroid must lie vertically between the two aponeurosis lines, discarding traces that sit outside the muscle proper. Each surviving component is reduced to its principal-axis line for the selection stage.

Table 4: Geometry engine parameters as shipped

Parameter	Value	Rationale
Apo component minimum size	500 px	Below this, fragments are noise

Parameter	Value	Rationale
Apo centroid separation	60 px	One band must not masquerade as two
Fascicle component minimum size	80 px	Removes speckle fragments
Candidate angle window	6 to 60 deg	Excludes aponeurosis-parallel junk and vertical artifacts
Candidate elongation	at least 4	Fascicles are lines, not blobs
Mode cluster half-width	3.5 deg	Fascicles within a muscle are near-parallel
Fascicles selected	top 3 by area	Mirrors the raters' 3-fascicle protocol
MT measurement sites	25 / 50 / 75 percent of width	Matches the 3-site averaging protocol
Consistency ratio FL sin(PA) / MT	0.4 to 2.5	Anatomically plausible triangle; real-mask median 1.04

5.3 Sin-weighted mode clustering: selecting the right fascicles

The selection stage answers the question the filters cannot: among the surviving candidates, which ones are actually the muscle's fascicles? The anatomical insight is that **fascicles within one muscle are roughly parallel**, so the true fascicles cluster tightly in angle while junk is spread. The engine therefore scans candidate angles to find the one that maximizes the sum of component areas weighted by the sine of the angle difference, effectively asking which angle has the most line-mass within a plus-or-minus 3.5 degree window, keeps the components within that window, and takes the top three by area as the selected fascicles.

The sine weighting is the non-obvious part and it exists because of a failure it fixed. An earlier version clustered by raw area, and large, flat, aponeurosis-adjacent junk kept winning the cluster vote, dragging the estimated angle toward zero. Weighting contributions by `sin(angle difference)` means a component only votes strongly for a cluster it crosses at a genuine angle, and near-horizontal mass contributes almost nothing. The three-fascicle top-area choice then mirrors the human reference protocol of Section 1.3, which averages three fascicles per rater, rather than trusting a single best line.

5.4 Computing FL and MT, and the consistency filter

Pennation angle is the mean angle of the three selected fascicle lines against the deep aponeurosis. Fascicle length is the mean, over selected fascicles, of the distance between the two intersection points where the fascicle's line, extended to infinity, crosses the two aponeurosis lines: because the computation uses lines rather than segments, **extrapolation beyond the field of view is inherent**, matching the definition in Table 1. Muscle thickness is the perpendicular distance between the two aponeurosis lines measured at 25, 50, and 75 percent of the image width and averaged, again matching the three-site protocol.

A final consistency filter guards the whole extraction: the ratio `FL x sin(PA) / MT`, which for a plausible fascicle-aponeurosis triangle should sit near one, must fall between 0.4 and 2.5. The bounds were calibrated on real masks, where the median ratio is 1.04, and the filter exists to catch the catastrophic case where the geometry has locked onto the wrong structures entirely, for example both aponeuroses from the same band or fascicles from an adjacent muscle. Images that fail the filter do not produce numbers at all; they fall through to the retry chain and, in the worst case, to the constants. A guard that occasionally discards a good image is cheaper than one that silently emits anatomy that is impossible.

5.5 Geometry ideas that failed

Four plausible upgrades to this engine were built, tested, and reverted, and they are recorded here in part to inoculate the future against retrying them. **4-connectivity labeling** shattered the one-pixel diagonal traces into fragments, producing the memorable bug where entire twin evaluations reported empty masks; one-pixel lines need 8-connectivity, full stop. The **vh filter**, which required fascicle candidates to span at least 35 percent of the muscle height, was designed against real masks, where fascicles do span the muscle, but predicted masks contain fragments rather than spanning lines, and the filter killed 121 of 122 twins, a perfect lesson in validating preprocessing against predicted rather than real masks.

Geometric constraint projection tried to improve FL by reconstructing it from MT and PA through the triangle identity, but a 2.5-degree PA error propagates through the identity into roughly an 11 mm FL error when PA is small, far worse than the direct line measurement, so the direct measurement won and the identity was demoted to the consistency filter where an error that large is a signal, not an output. Finally, **robust Huber line fitting** replaced least squares in an attempt to resist outlier pixels inside aponeurosis components; it improved the twins' PA bias by 0.13 degrees, looked encouraging, and then lost 0.011 on the leaderboard in submission v14, the third and final instance of the twin-positive-but-test-negative pattern that motivated the calibration doctrine of Section 6.3.

6. Validation Without Labels: The Twins Harness

6.1 Building evaluate_twins.py

The harness exists because of a budget problem: five submissions a day is far too slow a feedback loop to develop against, and the public leaderboard is a single noisy number that explains nothing. The 122 twins of Section 2.3 solved it. `evaluate_twins.py` runs the complete production chain, both segmentation models at their native resolutions, the full geometry engine, exactly as it would run on test images, on each twin, and then compares the result against reference values computed by running the same geometry engine on the **real**

masks with deliberately loose filters, so the reference is the best available stand-in for a human rater.

The output was designed for debugging, not just scoring. For each of PA, FL, and MT it reports MAE, mean bias, and correlation. It fits a linear shrinkage curve of predicted against reference PA, which is where the Chapter 8 correction constants came from. It lists the six worst images with their per-component angle breakdowns, which is how the dilation bias, the fold-2 poison, and the steep-angle blind spot were all first seen. And it renders QA overlays with real fascicles in green, predicted fascicles in red, and the two aponeuroses in cyan and magenta, which turned dozens of abstract metric movements into visibly diagnosable failures.

6.2 Standing marks

The best configuration's standing marks on the twins, stable across the final third of the campaign, are recorded in Table 5. The FL row is in pixels because the twins are evaluated before calibration, which is applied per test image and has no twin equivalent. The corrected PA row applies the Chapter 8 shrinkage correction to the raw PA prediction, and the improvement it shows, 2.96 down to 2.54 degrees, is the direct evidence on which that correction was adopted.

Table 5: Twins harness standing marks, best configuration

Metric	Value	Interpretation
Correlation (PA, reference vs predicted)	0.60	Solid for a derived-geometry pipeline
PA MAE, raw	2.96 deg	Within the 6-degree tolerance before any correction
PA MAE, after shrinkage correction	2.54 deg	The correction earns about 0.4 degrees
FL bias vs real-mask geometry	+37 px	Residual of the v3 dilation fix; was +110 px with v2
MT error vs real-mask geometry	approx. 24 px	Dominated by aponeurosis band placement

6.3 The calibration doctrine: twins generate hypotheses, the leaderboard arbitrates

The twins are not the test set, and the campaign learned to respect the difference through three expensive lessons. The twins are a single-muscle domain, predominantly the vastus-lateralis-like content of the fascicle pool, so they contain no rectus femoris, no gastrocnemius, and none of the handheld-device families. And they exaggerate errors by roughly a factor of two relative to the leaderboard, probably because twin reference values

come from loose-filter geometry on real masks rather than from human raters. The net effect is that twins evidence is directionally reliable but numerically untrustworthy.

Three specific cases established the pattern. Robust fitting improved twin PA by 0.13 degrees and cost 0.011 on the leaderboard. A refit of the PA correction on twin data improved the twins and lost on test. The classical structure-tensor blend of Chapter 10 improved twins and collapsed on the bipennate test families the twins do not contain. Against these, every change that the twins endorsed and that survived, the PA correction itself, the v3 models, the FL scale, the calibration direction, moved the leaderboard the predicted way. The doctrine that fell out is worth stating verbatim because it governed every decision after it:

> *The twins generate hypotheses; the leaderboard arbitrates constants. A change that improves the twins is a candidate for one submission, never a default. A change that improves the twins and has a mechanism that should generalize across muscles is a strong candidate. A change that improves the twins only through a constant is presumed guilty until the leaderboard confirms it.*

7. The Calibration Saga

7.1 Why calibration dominates the score

Two of the three targets, FL and MT, are lengths, and the pipeline computes them in pixels. Converting to millimetres requires a per-image scale factor, mm per pixel, and the arithmetic is unforgiving: a five percent error in the scale factor is a five percent error in every submitted FL and MT, regardless of how good the segmentation and geometry are. Because MT carries the tightest tolerance, three millimetres against typical values around twenty, calibration error is the most expensive kind of error in the competition. This is why calibration work ultimately contributed more score than any model generation, and why the largest single improvement of the campaign, the 0.19 points of submission v3, was purely a calibration fix.

The information needed to calibrate is burned into the images themselves: every device interface renders a depth ruler. The task was therefore to read those rulers, per family, as accurately as the pixels allow, and the two interface families required entirely different approaches.

7.2 The PNG family: labeled markers

The 58 PNG-family test images render their rulers as labeled tick markers at 0, 2, and 4 centimetres. Because the labels are burned-in text, the calibration is a direct measurement: locate the marker pair, measure the pixel distance between them, and set `mm per px = 20 / marker spacing`. The first working pipeline used a crude depth-based fallback instead, `depth_mm / 604`, which approximates the marker spacing from the nominal depth, and

replacing that fallback with the measured marker spacing was the whole content of submission v3, worth 0.192 in one step, the largest single gain of the campaign. The fallback remains in the code for images where marker detection fails, but in the final configuration it almost never fires.

7.3 The tif families: tick columns and borrowed priors

The 251 tif-family images are harder: their rulers are unlabeled tick columns, accompanied by interface text including the German settings words `De` and `Tiefe`. Without labels, absolute calibration has to be inferred from tick structure, specifically from the periodicity of major versus minor ticks and the known pixel dimensions of the two capture formats. The derived priors were **0.0656 mm per pixel for the 1200 by 800 format and 0.0830 for the 1088 by 644 format**. These constants carried the pipeline from submission v3 through v11b, and everything seemed consistent, until the bisection of Section 7.4 proved them five percent too high.

Table 6: Calibration constants by test family

Family	Images	Method	Constant
PNG	58	Labeled 2 cm markers, measured spacing	`20 / marker_px`, fallback `depth_mm / 604`
tif 1200 x 800	majority of 251	Tick structure prior	`0.0656 mm/px`, later x 0.95
tif 1088 x 644	remainder of 251	Tick structure prior	`0.0830 mm/px`, later x 0.95

7.4 The bisection: v12_m5 and the 0.95 correction

The suspicion that the tif priors were off came from cross-checking predicted MT distributions against the anatomical expectations built in Section 2.7: tif-family thicknesses ran systematically heavy. With five submissions a day available, the cleanest instrument was a **bisection**: submit the identical pipeline with only the tif scale factor changed, and let the leaderboard reveal the sign and rough magnitude of the error. Four submissions bracketed the answer in one day. Multiplying tif FL and MT by **0.95** scored 0.510, the best result of the campaign. Multiplying by 1.05 scored 0.681, confirming the direction of the error emphatically. Multiplying by 0.90 scored 0.583, overshooting the correction, and 0.97 scored 0.526, close but still short of 0.95's mark.

The pattern is textbook: 1.05 badly worse, 0.90 worse than 0.95 but much better than no correction, 0.97 slightly worse than 0.95, and 0.95 best. The true scale factor lies near 0.95, and the residual difference between 0.95 and the truth is smaller than the leaderboard's own

noise. The correction cost four submissions and bought 0.066 points, 0.576 down to 0.510, the single biggest late-stage gain, and it validated a principle the campaign relied on ever after: when a constant is suspect and submissions are affordable, a one-day bisection is the highest-information experiment available.

8. Post-Processing: Where a Third of the Score Came From

Between geometry and the submission CSV sits a thin layer of corrections: a pennation-angle transform, a fascicle-length scale, burst averaging, and range clipping. None of these touch a model or a filter; they are constants applied to derived numbers. Their combined contribution to the final score is on the order of a third of the total improvement, and the story of how each constant was found, validated, or rejected is the story of this chapter. The recurring pattern is the A/B pair: one submission with the change, one without or with its rival, decided in advance.

8.1 The pennation angle shrinkage correction

The twins harness measures not just PA error but the full relationship between predicted and reference PA, and that relationship was not the identity line: predicted angles regress toward the mean, underestimating steep fascicles and overestimating shallow ones, which is exactly what a noisy estimator does. The twins fitted a shrinkage line, and inverting it gives the correction the pipeline still applies today: **`PA = 7.54 + 0.59 x PA_raw`**.

The correction's adoption is the cleanest example in the campaign of a pre-declared A/B. Submission v9a applied a rival variant, mean-centered un-shrinkage, which uses the same slope idea but redistributes angles around the dataset mean instead of the fitted intercept. It scored 0.767 against the incumbent's 0.648, a decisive rejection. Submission v9b then applied the intercept form and scored **0.591 against the same 0.648 incumbent**, a decisive adoption. The verdict taught something subtle: the twins' fitted offset transfers to test, but the spread hypothesis behind the mean-centered variant does not. Shrinkage correction works; un-shrinking the spread is a different, wrong claim.

8.2 The fascicle length scale and the rejected coupling

Fascicle length carried a systematic overestimate whose roots were double: the +37-pixel dilation residue of Section 4.4 and residual calibration bias. The response was deliberately blunt, a global **`FL_SCALE = 0.925`** applied to every fascicle length. It entered as submission v6b, improving 0.713 to 0.685, and survived every subsequent comparison, including the v11-versus-v11b pair that tested 0.97 against 0.925 under the v3 models and confirmed 0.925 by 0.595 versus 0.576. A global scale is an admission that the bias is not yet

understood at the mechanism level, and that admission is honest: the alternative tried next was to understand it, and it failed.

The failed alternative was **sin-coupling**, submission v10. The hypothesis was attractive: since FL relates to MT through the pennation triangle, `FL = MT / sin(PA)`, a systematic PA error should produce a predictable FL error, so correcting FL by the ratio of sines of corrected and raw PA should beat a global scale. The twins loved it. The leaderboard scored it 0.667 against the incumbent's 0.591, a rejection costing one submission and settling the question: **FL error is per-image noise, not angle-coupled bias**. The triangle identity governs the consistency filter, where violations indicate broken geometry, but it does not govern the error statistics, and the two facts are compatible because the dominant FL errors are selection and endpoint errors, not PA-propagation errors.

8.3 Burst averaging and range clipping

The last two post-processing steps are guards rather than corrections. **Burst averaging**, justified by Section 2.6, replaces each image's prediction with the mean over its near-duplicate run at correlation above 0.99, converting 28 bursts of correlated frames into 28 single low-variance measurements. It never had a dedicated A/B because it is nearly free variance reduction with no mechanism for harm, and it survived every configuration it shipped inside. **Range clipping** then bounds every output to anatomically plausible spans, PA in 5 to 45 degrees, FL in 30 to 200 mm, MT in 10 to 50 mm. Clipping never changes a healthy prediction; its entire job is to cap the damage when the fallback chain has been reached or the geometry has locked onto something wrong, and the constants row of Table 11 records the exact bounds.

9. The Complete Submission Log

Twenty submissions were made in total. Table 7 lists every one of them with its content and score, and Figures 1 and 2 show the trajectory and the breakdown of where the improvement came from. The narrative below groups them into four phases, because the phases have distinct characters: exploration, grind, correction, and endgame. One reading of the table that deserves emphasis before the detail: of twenty submissions, only eight hold or improve the running best; the other twelve were experiments whose value was information, and most of that information is now Chapter 10.

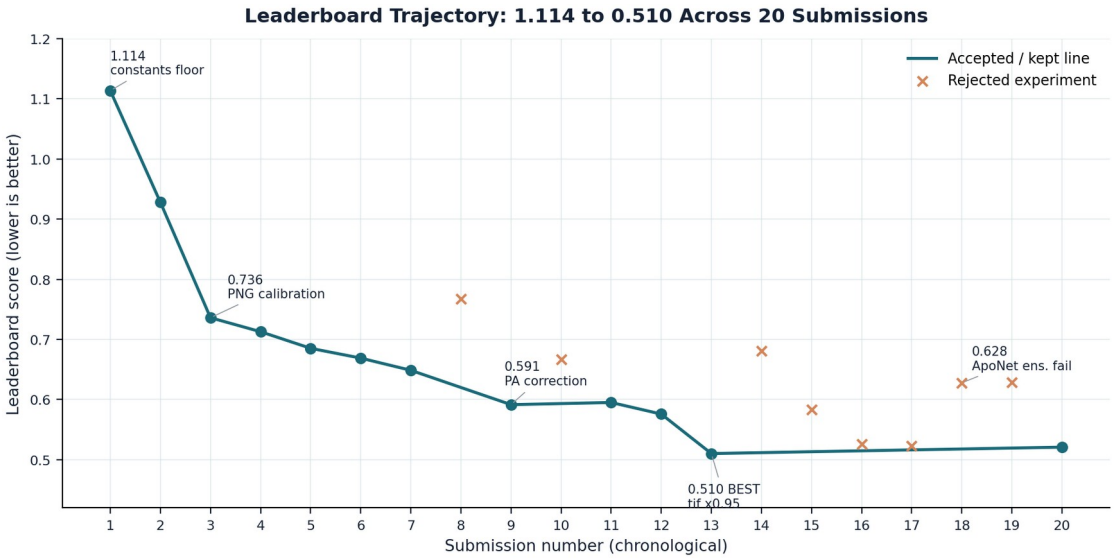


Figure 1: Leaderboard score across all 20 submissions. The line traces the running-best chain; crosses mark rejected experiments.

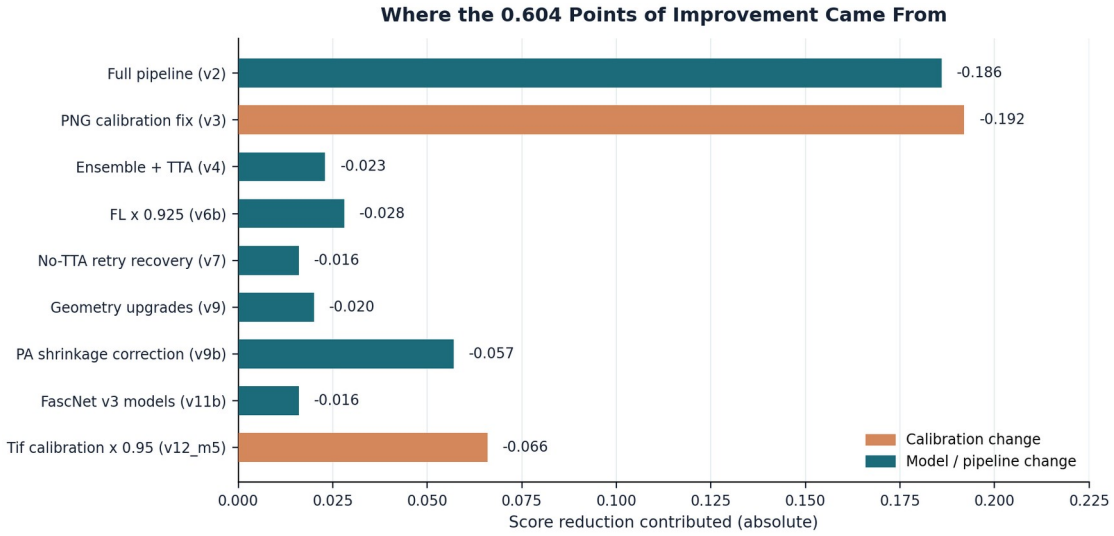


Figure 2: Score reduction contributed by each accepted change. Calibration changes (warm color) dominate.

Table 7: All twenty submissions, in order

#	Submission	Change	Score	Verdict
1	v1 constants	Constants 18.0 / 80.0 / 23.0; format and floor anchor	1.11370	Floor established
2	v2	First full pipeline end to end	0.928	Adopted
3	v3	PNG calibration from measured markers	0.736	Adopted, +0.192
4	v4	3-fold v2 ensemble plus TTA	0.71269	Adopted, marginal
5	v6b	FL x 0.925	0.68508	Adopted
6	v7	No-TTA retry recovery in fallback chain	0.66877	Adopted
7	v9	Geometry: 6-deg floor, sin-mode clustering, closing	0.64848	Adopted
8	v9a	PA mean-centered un-shrink	0.76693	Rejected

#	Submission	Change	Score	Verdict
9	v9b	PA = 7.54 + 0.59 x PA	0.59136	Adopted, +0.057
10	v10	FL sin-coupling correction	0.66694	Rejected
11	v11	v3 models, FL 0.97	0.59497	Neutral
12	v11b	v3 models, FL back to 0.925	0.57566	Adopted
13	v12_m5	tif FL and MT x 0.95	**0.510**	**Best, adopted**
14	v12_p5	tif x 1.05 (bisection)	0.68067	Rejected
15	v12_m10	tif x 0.90 (bisection)	0.58334	Rejected
16	v12_m3	tif x 0.97 (bisection)	0.5257	Rejected
17	v14	Robust fit, inline calibration, PA refit	0.52258	Rejected
18	v15	ApoNet 2-seed ensemble	0.62764	Rejected, catastrophic
19	v15_blended	Plus classical PA blend	0.62843	Rejected
20	v16	Revert to single ApoNet	0.52067	Partial recovery

9.1 Phase one: the floor and the first pipeline (v1 to v4)

Submission one was deliberately content-free: the constants 18.0 degrees, 80.0 mm, and 23.0 mm for every image. Its purposes were to prove the CSV format, comma-separated with 309 rows asserted in-script, and to anchor the floor at 1.114, and it served both. Submission two put the first complete pipeline on the board at 0.928, a 0.186 improvement that mostly proved the pipeline runs; its calibration was still the crude depth-based fallback. Submission three, the PNG marker fix, was the campaign's single biggest gain at 0.192 and established calibration as the dominant lever. Submission four, the v2 ensemble with TTA, added a marginal 0.023, and its modesty, in hindsight, was the first evidence for the correlated-folds lesson of Section 4.5.

9.2 Phase two: the post-processing grind (v6b to v9)

Phase two was three consecutive single-constant wins: FL times 0.925 worth 0.028, the no-TTA retry recovery worth 0.017, and the geometry package of submission v9, the 6-degree floor plus sin-mode clustering plus binary closing, worth 0.020. None was individually dramatic, but together they took the running best from 0.713 to 0.648, and each was a one-variable change validated against its predecessor, the discipline that made the phase productive. The retry-recovery change deserves its one sentence of explanation: v7 removed test-time augmentation from the fallback path, having observed that when the primary model failed on a hard image, the TTA-averaged retry often produced something worse than the un-augmented one, and the recovery path should degrade toward simplicity, not complexity.

9.3 Phase three: the correction era (v9b to v12_m5)

Phase three contains both the sharpest rejection and the sharpest adoption of the campaign, back to back. The v9a-versus-v9b pair of Section 8.1 spent two submissions to settle the PA correction and earned 0.057. The v10 rejection of Section 8.2 spent one to settle that FL error is noise. The v11-versus-v11b pair then integrated the v3 models, with FL 0.97 scoring 0.595 against FL 0.925's 0.576, confirming that the v3 models improved the pipeline without changing the optimal FL scale. The phase closed with the bisection quartet of Section 7.4: v12_m5's 0.510, the campaign best, bracketed by three submissions whose only job was to prove that 0.95, not 0.97 or 0.90, was the right correction. Phase three took the running best from 0.648 to 0.510 in ten submissions.

9.4 Phase four: the endgame failures and recovery (v14 to v16)

Phase four is three ideas, three rejections, and a revert. Submission v14 bundled three twin-positive refinements, robust Huber fitting, inline calibration, and a twin-refit PA correction, and scored 0.523, slightly worse than the incumbent: the first clear demonstration that the twins' constants had been squeezed dry. Submission v18's ApoNet ensemble then went catastrophically wrong at 0.628, a 0.107 loss, the campaign's worst single result and the subject of Section 4.6. Submission v19 added the classical blend to the ensemble and matched the damage at 0.628, and Section 10.3 explains why its failure mode, the bipennate collapse, was invisible to the twins. Submission v20 reverted to the single ApoNet and recovered to 0.521, within noise of the best, closing the campaign exactly where the doctrine of Section 6.3 said it should: **the best configuration is v12_m5 at 0.510, and the endgame experiments confirmed rather than displaced it**.

10. The Negative Results Catalog

Fifteen distinct ideas were built or designed, tested, and rejected over the campaign. They are cataloged in Table 8 with their failure reasons, because a negative result that is recorded is a permanent asset: it is the boundary of the solution space, and it protects the future from re-spending submissions on the same hypothesis. Three of the fifteen taught lessons deep enough to deserve their own paragraphs below the table.

Table 8: Negative results, in rough order of discovery

Idea	Where tested	Outcome	Why it failed
4-connectivity labeling	Twins	Bug	Shatters 1 px diagonal lines; empty twin masks
vh filter, 35 percent muscle height	Twins	Killed 121/122	Designed against real masks; predictions are fragments
Geometric constraint projection for FL	Twins	Rejected	2.5-deg PA error becomes ~11 mm FL; worse than direct
FL sin-ratio coupling	LB v10	0.667 vs 0.591	FL error is per-image noise, not angle-coupled bias

Idea	Where tested	Outcome	Why it failed
PA mean-centered un-shrink	LB v9a	0.767 vs 0.648	Offset transfers; spread hypothesis does not
Correlated-fold ensembling	LB v4	Marginal	Same data and recipe yields near-identical models
Hot mid-schedule checkpoints	Twins	Rejected	Spiky probabilities poison ensemble averages
Wrong-resolution evaluation	Twins	Bug	Each model must predict at its training resolution
Plus-minus 30-deg rotation augmentation	Twins	Rejected	Augmentation must match test range; corr 0.60 to 0.38
ApoNet 2-seed ensemble	LB v15	0.628 vs 0.521	Consensus behaves differently than either member
Robust Huber line fitting	LB v14	0.523 vs 0.510	Twin-positive, test-negative; constants exhausted
Classical structure-tensor PA blend	LB v19	0.628	Bipennate RF: opposing fans cancel in circular mean
Distillation regressor (ResNet50)	Twins	4.29 deg PA	Imitation of a pipeline is not improvement on it
Test-image pseudo-label fine-tuning	Design review	Not run	Would reinforce regression-to-mean under domain shift
Anisotropy FL correction x 0.85	A/B	Neutral	Raters used isotropic vertical scale

10.1 The ApoNet ensemble: a consensus that was worse than either member

The most expensive entry in the table is also the most theoretically interesting. Two ApoNet seeds, each individually fine by every local metric, probability-averaged into a consensus that cost 0.107 on the leaderboard. The mechanism, reconstructed afterward, is that the two models disagree most on exactly the hard images, the domain-shift families, where each model's probability map is uncertain in a different way; averaging produces a diffuse, lower-confidence mask whose thresholded shape is worse than either parent's, and the geometry engine, which depends on band edges for its polynomial fits, converts slightly softer bands into measurably worse lines. The general lesson extends beyond ensembles: **any change to a segmentation model must be evaluated end to end through geometry, because mask-space metrics do not predict geometry-space outcomes**.

10.2 The classical blend and the bipennate blind spot

The classical structure-tensor blend was motivated correctly, the pipeline's learned PA has a shallow bias, so blend in an image-processing estimate of dominant orientation as a second opinion, but the estimator was chosen wrong. The structure-tensor dominant orientation, aggregated with a circular mean, presumes a single dominant direction in the region. On the bipennate rectus femoris families, 20 through 24 of Section 2.7, fascicles fan in **opposite directions** from a central aponeurosis, and two equal and opposite fans cancel in the circular

mean, collapsing the estimate toward horizontal, precisely the shallow bias the blend was meant to correct. The twins, being vastus-lateralis-like, unipennate, contain no such anatomy, which is why the blend looked good locally and destroyed the bipennate families on the leaderboard. The failure is the single best argument in the campaign for the doctrine of Section 6.3: validation data that lacks a domain cannot veto failures in that domain.

10.3 The distillation regressor: imitation is not improvement

The regressor experiment answered a question that kept recurring: since the pipeline produces numbers, why not train a network to produce those numbers directly, faster and end to end? A ResNet50 with three heads was trained on pipeline pseudo-labels and fit them to 0.87 degrees of PA error, a number that would have looked excellent in a results table. On the twins, it produced 4.29 degrees, several times the pipeline's own error. The resolution of the paradox is that the regressor learned the pipeline's average behavior as a function of global image appearance, discarding the per-image geometric evidence, and its errors were therefore largest exactly where the pipeline's evidence mattered most, the hard images. Global regression is domain-sensitive in a way that local segmentation is not, and the experiment closed the question of a regression shortcut permanently, at the cost of one training run and zero submissions.

11. The Final Architecture and How to Reproduce It

The best-scoring configuration, v12_m5 at 0.510, is a completely determined pipeline: fixed checkpoints, fixed constants, fixed geometry, plus one post-generation CSV transform. This chapter records it as a stage table and a replication procedure, so that the result is reproducible from the artifacts alone. Table 9 walks the pipeline in execution order; Table 11 in Appendix B collects every constant in one place.

Table 9: The v12_m5 pipeline, stage by stage

Stage	Component	Configuration
Fascicle masks	FascNet v3, folds 0 and 1	640 px inputs, probability averaging, horizontal-flip TTA, threshold 0.5, binary closing
Aponeurosis mask	ApoNet v1	512 px inputs, threshold 0.5
Geometry	Engine of Chapter 5	Table 4 parameters; consistency ratio 0.4 to 2.5
Calibration	Per image	PNG: 20 / marker px; tif: 0.0656 or 0.0830 by format
PA correction	Shrinkage inverse	PA = 7.54 + 0.59 x PA_raw
FL scale	Global	FL x 0.925
Burst averaging	Correlation runs	Mean over r > 0.99 runs
Clipping	Range guards	PA 5-45, FL 30-200, MT 10-50
CSV transform	Post-generation	tif-suffixed rows: FL and MT x 0.95

Stage	Component	Configuration
Fallback chain	Degradation path	v3 ensemble, then v1 + TTA, then v1, then constants 18.0 / 80.0 / 23.0

The replication procedure, from the project root with the artifacts of Appendix A in place, is four steps. Each step consumes the named artifacts of the previous one, and the final assertion of 309 rows in the submission script is what catches any silent upstream failure before it reaches the leaderboard.

1. Run the forensics suite to regenerate `content_hashes.csv`, `twins.csv`, `fasc_mask_inventory.csv`, and `calibration_v1.csv`, and verify the pool counts and the 122 twins against Chapter 2.
2. Train ApoNet v1 on the 888 clean aponeurosis images and FascNet v3 folds 0 and 1 at 640 pixels with dilation-1 targets, verifying ApoNet against the twins at a Dice of about 0.82.
3. Run `infer_submit.py` with the Chapter 5 geometry, the Chapter 8 corrections (`PA_A` 7.54, `PA_B` 0.59, `FL_SCALE` 0.925), burst averaging, and clipping, with the fallback chain armed.
4. Apply the CSV transform, multiplying FL and MT by 0.95 on tif-suffixed rows, assert 309 rows, and submit.

12. Infrastructure Lessons

The project ran on a Windows machine with an RTX 5050 laptop GPU, and the environment fought back in ways that cost real hours before they were understood. None of the items in this chapter are about machine learning; all of them are about the gap between documentation and reality, and every one of them produced a standing rule in the codebase. They are recorded here because a replication attempt without them will hit the same walls.

The GPU is newer than the default wheels. The RTX 5050 is a Blackwell part, compute capability `sm_120`, and the default PyTorch builds do not contain kernels for it, failing with the cryptic error `no kernel image is available for execution on the device`. The fix is installing PyTorch from the CUDA 12.8 index, `--index-url https://download.pytorch.org/whl/cu128`. This was the first wall and the most confusing, because training code runs fine on CPU, so the failure only appears on the first GPU forward pass.

DLL load order is load-bearing. On this Windows setup, importing `torch` after `numpy`, `scipy`, or `PIL` triggers `WinError 1114` on `c10.dll`, a DLL initialization failure caused by conflicting runtime DLLs already loaded into the process. The rule that eliminated an

entire class of crashes: `import torch` must be the first import in every entry-point script, before anything else, with `KMP_DUPLICATE_LIB_OK=TRUE` set in the environment to defuse the OpenMP runtime duplicate. **DataLoader workers need the spawn guard**: Windows uses spawn semantics for child processes, so any script using `num_workers` above zero requires the `if __name__ == '__main__':` guard around the entry point, or the workers re-import the script and hang; the alternative used in several forensics scripts was simply `num_workers=0`.

Two final small items round out the list. The submission format is **comma-separated** `image_id,pa_deg,fl_mm,mt_mm`; the sample submission file's semicolons are a locale artifact of the machine that exported it, and the local verification of the official scoring code is what settled the question, which is why every submission script asserts 309 rows and a parseable header before writing. And a Python version detail that bit the code generation more than once: **f-strings cannot contain backslashes in their expressions before Python 3.12**, so path-building expressions inside f-strings had to be hoisted to variables, a trivial rule that nonetheless produced several confusing syntax errors before it became habit.

13. The Unbuilt Roadmap

When the campaign ended, four ideas remained on the board, evaluated but unbuilt. They are recorded here with their assessments, both because they are the natural next steps for anyone continuing the work and because the reasoning about their expected value is itself part of the project record.

The fascicle endpoint network is the only remaining idea with an estimated upside above 0.05. The concept is to replace the segmentation-plus-selection-plus-fitting chain for fascicles with a direct keypoint model that localizes the six points, fascicle origin and insertion on each of three fascicles, and computes PA and FL from the points directly. The motivation is that the current pipeline's dominant errors are selection and endpoint errors, which the Chapter 8 corrections patch globally but never cure; keypoint localization attacks them at the source. The training signal exists: the mask pool's traced fascicles provide endpoints wherever they meet the aponeuroses, and the twins would validate the geometry identically. The risk is the same one that killed the distillation regressor, that direct regression is domain-sensitive, but keypoints are local predictions, which is the property that made segmentation transfer well.

External labeled data was identified and then parked. The GasFasTrack 2019 dataset contains manually measured fascicle lengths and pennation angles from a Telemed device on five subjects, genuinely labeled values rather than derived ones, the only such data identified in the whole project. It was parked on a 30-gigabyte download that the campaign never found room for, and the one preparatory step that would have been cheap, hashing its image list

against the training pools to check for overlap, was never run. The realistic assessment is modest: five subjects is narrow, and Telemed is one device among the 29 families, but it would have provided the only external anchor on absolute PA and FL truth the project ever had access to.

The final retrain with twins included is the classic endgame move: fold the 122 twins, the highest-quality completely-labeled data in the project, back into training for the final model generation, accepting the loss of the validation harness in exchange for 6.6 percent more of the best data. It was deliberately saved for last, because once done it cannot be undone, and the endgame experiments of Section 9.4 never reached the point where spending the harness made sense. **Rectus-femoris-specific geometry** is the last item: a three-band detection mode and outermost-pair fascicle-length computation for the bipennate families, addressing the anatomy the classical blend failed on. It remains attractive precisely because it is anatomical rather than statistical, which is the profile of every change that worked in this campaign.

14. Meta-Lessons from the Whole Campaign

Strip away the anatomy and the ultrasonography, and the campaign is a case study in making decisions under a expensive feedback budget, and the lessons at that level generalize further than any single result. Five of them emerged, each one earned by at least one submission, and they close the chronicle because they are the part of the project that transfers.

Instrument before submitting. The 122 twins cost zero submissions and predicted the direction of nearly every leaderboard verdict of the campaign, including the rejections. The single highest-leverage act of the whole project was building the harness before building the second model, and the counterexamples, the three twin-blind failures of Section 6.3, were failures of domains the harness did not contain, not failures of the harness principle. The general form: before optimizing against an expensive oracle, build a cheap oracle, learn its calibration against the expensive one, and spend accordingly.

Post-processing has a ceiling. Calibration and correction bought the campaign 1.114 down to 0.51, more than half of the total improvement, without touching a model weight after submission four. Then the ceiling arrived, announced by three consecutive twin-positive rejections, and every attempt to push past it confirmed it. The general form: when derived constants have been fitted to the available signal, further gains must come from the model or the data, and recognizing that moment early saves a week of grinding.

Constants transfer directionally, not numerically. Every twin-measured constant that helped on the leaderboard helped in the twins first, and no constant that helped in the twins transferred at face value: the shrinkage intercept transferred, its spread did not; the robust-fit bias improvement inverted; the FL scale was measurable but its optimum differed by three

percent. The general form: a proxy metric is a compass, not a ruler, and A/B structures must be designed with the compass in mind.

Single-variable A/Bs with pre-declared decision rules. Every rejection in Table 8 cost one submission and produced permanent knowledge, and every one of them was a single change with a verdict rule agreed before the result arrived. The discipline matters most when a result is ambiguous, and the bisection of Section 7.4 is the purest example: four submissions, each differing by one multiplier, producing a bracketed answer in a single day. The general form: an experiment that cannot change a decision is not worth a submission, and an experiment that changes two things cannot be attributed.

Domain knowledge beats generic machine learning. The two biggest wins of the campaign, the depth-ruler calibration and the muscle map, were anatomy and physics, not modeling; the two biggest failures, the bipennate collapse and the distillation regressor, were generic techniques applied where domain structure contradicted their assumptions. The general form, and the sentence that would headline the post-mortem if only one sentence survived: understand what the data actually contains, in the most literal sense, before choosing what to learn from it.

> *Final standing: 0.510, submission v12_m5, from a 1.114 floor. Best configuration unchanged since the bisection. The twins remain excluded from training, the roadmap remains open, and every artifact needed to resume is cataloged in Appendix A.*

Appendix A: Artifact Manifest

Everything the campaign produced lives in the project root and is named in Table 10 by category. Checkpoints are PyTorch state dictionaries; dataframes are CSV files consumed by the scripts that name them; the submissions are the exact CSVs sent to the leaderboard. The two failed ApoNet ensemble seeds are retained deliberately, as the reproducible artifact of Section 10.1's most expensive lesson.

Table 10: Artifacts by category

Category	Artifacts
Checkpoints	`aponet_v1.pt`; `fascnet_v1.pt`; `fascnet_v3_fold0.pt`, `fascnet_v3_fold1.pt` (fold 2 broken, retained); `fascnet_v2_fold0/1/2.pt`; `regressor_v1.pt`; `aponet_ens_0.pt`, `aponet_ens_1.pt` (failed ensemble)
Dataframes	`content_hashes.csv`; `twins.csv`; `fasc_mask_inventory.csv`; `calibration_v1.csv`; `twin_eval.csv`; `pseudo_labels.csv`; `thumbs32.npz`; `regressor_stats.csv`
Forensics scripts	`data_analysis.py` and `data_analysis_2.py` (pools, duplicates, twins); `dd_calibration.py` (depth-ruler derivation)

Category	Artifacts
Training scripts	`training_seg.py` (ApoNet); `train_fascnet_v2.py`, `train_fascnet_v3.py`, `train_fascnet_v4.py` (rotation experiment); `train_aponet_ens.py`; `train_regressor.py`
Pipeline	`infer_submit.py` (production inference and submission generation, fallback chain armed)
Validation	`evaluate_twins.py` (harness); `diag_geo.py` (geometry diagnostics); `eval_regressor_twins.py`; `classical_pa.py` (blend experiment)
Transforms	`gen_pseudo_labels.py`; `apply_classical_blend.py`; `make_v9ab.py` (A/B pair generation)
Submissions	`submission_v2.csv` through `submission_v16.csv`, plus A/B variants

Appendix B: Final Constants and Parameters

Table 11 collects every constant of the v12_m5 configuration in one place, as the single reference for replication or audit. Constants are grouped by the chapter that justified them, and the values are exactly those in the shipped code, with no rounding beyond what the code itself contains.

Table 11: Complete constants of the v12_m5 configuration

Constant	Value	Justified in
ApoNet input resolution	512 px	Section 4.1
FascNet v3 input resolution	640 px	Section 4.4
FascNet target dilation	1 px	Section 4.4
Ensemble method	probability averaging, native resolutions	Section 4.4
TTA	horizontal flip, primary path only	Section 9.2
Mask threshold	0.5	Section 4.1
Morphology	binary closing	Submission v9
Apo component minimum / separation	500 px / 60 px	Section 5.1
Fascicle component minimum	80 px	Section 5.2
Candidate angle window	6 to 60 degrees	Section 5.2
Candidate elongation	at least 4	Section 5.2
Mode cluster half-width	3.5 degrees	Section 5.3
Fascicles selected	top 3 by area	Section 5.3
MT sites	25 / 50 / 75 percent of width	Section 5.4
Consistency ratio bounds	0.4 to 2.5	Section 5.4

Constant	Value	Justified in
PNG calibration	`20 / marker_px`, fallback `depth_mm / 604`	Section 7.2
tif calibration priors	`0.0656` (1200x800), `0.0830` (1088x644) mm/px	Section 7.3
tif correction	FL and MT x 0.95	Section 7.4
PA correction	`PA = 7.54 + 0.59 x PA_raw`	Section 8.1
FL scale	x 0.925	Section 8.2
Burst threshold	r > 0.99	Section 8.3
Clipping ranges	PA 5-45, FL 30-200, MT 10-50	Section 8.3
Fallback constants	18.0 deg / 80.0 mm / 23.0 mm	Section 3